



AEGENT

Security Assessment

Assessment dated Jul 30th, 2026

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SUMMARY

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Executive Summary

AAGENT

Smart-contract security assessment

TYPE

DeFi / token infrastructure

ECOSYSTEM

BNB Chain (BSC)

METHODS

Manual analysis, static analysis, testing

LANGUAGE

Solidity 0.8.24

TIMELINE

Completed 30 July 2026

CODEBASE

Scope root 5c51c8e5df9126e3b0...

Vulnerability Summary

0 **Critical**
No Critical findings

0 **High**
No High findings

4 **Medium**
4 Resolved

1 **Low**
1 Resolved

4 **Informational**
4 Resolved

9 **Total Findings**

9 **Resolved**

All tracked code findings verified

Production activation remains conditional on deployment-address, role, reserve, RPC and funded-wallet acceptance gates.

Vulnerability Summary

0 CRITICAL · 0 HIGH · 4 MEDIUM RESOLVED · 1 LOW RESOLVED · 4 INFORMATIONAL RESOLVED

CRITICAL / HIGH

0 Critical and 0 High findings.

MEDIUM

4 Medium findings tracked; all code remediation verified.

LOW / INFORMATIONAL

1 Low and 4 Informational findings tracked; all remediation verified.

RESIDUAL ACTIVATION RISK

Production deployment evidence is intentionally outside the code snapshot and remains a release gate rather than an open code vulnerability.

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REPORT STRUCTURE

Executive Summary; Vulnerability Summary; Codebase; Audit Scope; Approach & Methods; Assessment Notes; Findings; Verification; Deployment Controls; Appendices; Disclaimer.

NAVIGATION

Every page carries the institution identity, AEGENT project identity and an explicit page number.

CODEBASE

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Codebase

LANGUAGE AND ECOSYSTEM

Solidity 0.8.24, Hardhat 2.29.0, OpenZeppelin Contracts 5.1.0, BNB Chain mainnet chain ID 56.

CORE CONTRACTS

AagentMarketRegistry, AagentSwapV2, AagentRedemptionV2, AagentSaleProceedsVault and AagentStakingV1.

SOURCE IDENTITY

Scope root SHA-256: 5c51c8e5df9126e3b0a5ce15e98869f6130e43bb13e4caaebf30853286763f0c.
Evidence run: 20260730T020710Z.

Audit Scope

INCLUDED

Market configuration, phase pricing, purchases, redemption cost basis, reserves, oracle safety, staking custody, ownership and deployment controls.

EXCLUDED

Third-party token implementations, live RPC availability, exchange listing, production key custody and any address not bound to the printed scope root.

CHANGE POLICY

Any byte-level change to an in-scope file requires a new scope root and regression execution.

SCOPE CHECKSUMS

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Scope Checksums

ROOT BINDING

Production Solidity scope root:

5c51c8e5df9126e3b0a5ce15e98869f6130e43bb13e4caaebf30853286763f0c.

EVIDENCE BINDING

Frozen evidence run: 20260730T020710Z; 35 frozen inputs; 11 production Solidity files.

INTEGRITY

Relative path and lowercase SHA-256 records are sorted and joined with LF before the root digest is computed.

Approach & Methods

MANUAL ANALYSIS

Line-by-line control-flow, access-control, accounting, oracle, custody, replay, reentrancy, precision and denial-of-service analysis.

AUTOMATED ANALYSIS

Hardhat compilation and tests, solidity-coverage, Slither 0.11.5 and production dependency advisory checks.

ADVERSARIAL SCENARIOS

Stale quotes, stale or future oracle rounds, reserve shortfall, split redemption, nonce replay, wrong chain, reentrancy and fee-on-transfer assets.

Assessment Notes

TRUST MODEL

The Registry is the authoritative schedule and market-mode source. Swap, Redemption and Staking bind to explicit endpoints and fail closed on inconsistent facts.

PRIVILEGED ACTIONS

Owner controls remain bounded by pause states, inventory/reserve locks, endpoint precommitment and non-renounceable ownership.

OPERATIONAL DEPENDENCY

Deployment must bind audited bytecode, verified addresses, production feeds, custody roles and funded reserves before enablement.

Privilege Analysis

OWNER

Configuration is owner-restricted; ownership cannot be renounced.

PUBLIC KEEPER

Any keeper may persistently pause unhealthy purchase feeds, but cannot disable redemption-only modes.

ECONOMIC INVARIANTS

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Economic Invariants

SUPPLY AND CAPS

Global sale cap, limited-window inventory and per-wallet limits are enforced on chain.

REFUND LIABILITY

Projected purchases cannot exceed verified redemption reserves.

Oracle Risk Model

INPUTS

BNB/USD, USDT/USD and USDC/USD feeds have independent age, bound and deviation settings.

RECOVERY

Risk parameters update only while purchases are paused.

Redemption Accounting

BASIS

Wallet cost basis survives partial refunds and phase changes.

REPLAY

Each exact redemption consumes one wallet execution nonce.

STAKING CUSTODY

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Staking Custody

ENTITLEMENT

Position stores asset, principal, USD value, term, reward and maturity.

RELEASE

Beneficiary proceeds release only after the matured AGNT claim.

EVENT EVIDENCE

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Event Evidence

CANONICAL FACTS

Transaction and event identity are verified before settlement.

FINALITY

Block-hash drift, confirmation rollback and duplicate evidence fail closed.

TOOLCHAIN VERSIONS

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Toolchain Versions

COMPILER

solc 0.8.24; optimizer 500; EVM paris.

ANALYSIS

Hardhat 2.29.0; solidity-coverage 0.8.17; Slither 0.11.5.

Source File Register

CORE

Five production core contracts.

INTERFACES

Six production interfaces.

PACKAGE

Thirty-five frozen inputs including tests, scripts and configuration.

ARCHITECTURE

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Architecture

REGISTRY

Authoritative 90-day schedule, fixed 10/7/3 phase rates, five market modes, endpoint precommitment and configuration versioning.

SWAP AND REDEMPTION

Purchases validate quotes, phases, oracles, caps and reserves; refunds use actual wallet cost basis and one-time execution nonces.

VAULT AND STAKING

Sale proceeds remain locked; staking records full entitlement, validates value and releases proceeds only after claim.

Trust Boundaries

USER BOUNDARY

Wallets sign exact calldata; replay, stale quotes and mismatched value are rejected.

ORACLE BOUNDARY

Round completeness, timestamps, maximum age, absolute bounds and deviation limits are validated on chain.

OPERATIONS BOUNDARY

Keeper and deployment tooling reject missing configuration, wrong chain and secret-bearing public artifacts.

MED-01 - Cumulative redemption threshold

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with cumulative redemption threshold could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

Wallet-level cumulative UTC-day accounting now evaluates the next total before the hard limit and manual threshold.

VALIDATION

Targeted regression tests pass for split transactions, threshold equality and hard-limit enforcement.

MED-02 - Redemption reserve coverage

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with redemption reserve coverage could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

BNB, USDT and USDC purchase execution now fails closed unless the refund reserve covers projected cost-basis liability.

VALIDATION

Fresh tests cover reserve readability, liability growth and insufficient-reserve rejection.

MED-03 - Persistent oracle circuit breaker

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with persistent oracle circuit breaker could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

Per-feed maximum age, absolute bounds, deviation limits and permissionless persistent purchase pause are implemented.

VALIDATION

Breaker reason classification, healthy-feed rejection and post-launch redemption continuity tests pass.

MED-04 - Staking oracle and operational controls

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with staking oracle and operational controls could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

Staking validates round completeness, timestamps, decimals, asset/feed collisions, quote lifetime, slippage and inventory.

VALIDATION

Constructor, quote, custody, reentrancy and keeper safety suites pass; production activation remains gated by deployment controls.

LOW-01 - Redemption rate-limit semantics

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with redemption rate-limit semantics could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

The enforced cumulative daily wallet limit and manual threshold prevent transaction splitting and bound wallet throughput.

VALIDATION

Boundary tests pass; the chosen UTC-day policy is documented as the intended control.

INFO-01 - Event evidence completeness

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with event evidence completeness could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

Canonical transaction, event, nonce, position and block-hash evidence is persisted for settlement workflows.

VALIDATION

Duplicate and drift conditions fail closed in the service verification layer.

INFO-02 - NatSpec and interface clarity

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with natspec and interface clarity could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

External interfaces and security-sensitive configuration fields are documented and version-bound.

VALIDATION

Compilation succeeds across 34 Solidity files with the current interfaces.

INFO-03 - Deployment reproducibility

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with deployment reproducibility could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

The deployment script validates the exact 90-day schedule, chain 56, feed constraints, predicted addresses and constructor events.

VALIDATION

Eight deployment-safety regression cases pass.

INFO-04 - Operational runbook controls

STATUS - RESOLVED

STATUS

RESOLVED

RISK

The earlier control gap associated with operational runbook controls could weaken enforcement under adversarial sequencing or dependency failure.

REMEDIATION

Keeper configuration, irreversible broadcast confirmation and public-artifact secret stripping are encoded as fail-closed checks.

VALIDATION

Keeper and deployment tests reject missing, malformed, testnet and secret-bearing inputs.

Finding Closure Matrix

9 OF 9 TRACKED FINDINGS RESOLVED

MEDIUM

MED-01, MED-02, MED-03 and MED-04: RESOLVED in the current code snapshot.

LOW

LOW-01: RESOLVED with documented UTC-day cumulative wallet policy.

INFORMATIONAL

INFO-01 through INFO-04: RESOLVED through evidence, documentation and deployment-control hardening.

COMPILATION RESULTS

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Compilation Results

COMPILATION · PASS**COMPILER**

Solidity 0.8.24, optimizer enabled with 500 runs, EVM target paris.

RESULT

34 Solidity files compiled successfully with zero compiler errors.

DEPENDENCIES

OpenZeppelin Contracts 5.1.0 and pinned Hardhat plugins.

Test Results

TESTS · PASS

CHAIN 56

74 passing tests; one intentional wrong-chain constructor case is skipped by design under the chain-56 configuration.

CHAIN 97

The independent wrong-chain constructor guard passes 1/1.

COVERAGE AREAS

Registry, purchase, redemption, reserve, staking, deployment and keeper safety.

COVERAGE RESULTS

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Coverage Results

PRODUCTION CONTRACTS

96.02% statements; 59.30% branches; 97.09% functions; 86.25% lines.

ALL INSTRUMENTED FILES

96.18% statements; 59.91% branches; 97.66% functions; 87.01% lines.

INTERPRETATION

Coverage supports execution evidence but is not a guarantee of exploit absence.

Static Analysis

STATIC ANALYSIS · COMPLETE

TOOL

Slither 0.11.5 completed successfully.

TRIAGE

71 raw detector records were manually classified against the system design and tracked findings.

RESULT

No unresolved Critical or High issue remained after source-level validation.

DEPENDENCY AUDIT

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Dependency Audit

DEPENDENCY AUDIT · PASS**PRODUCTION ADVISORIES**

npm audit --omit=dev reported zero known production dependency advisories.

VERSION CONTROL

Security-sensitive dependencies and tool versions are pinned in the frozen package.

Deployment Script Safety

CHAIN AND INTENT

Requires chain 56 and an exact irreversible broadcast confirmation phrase.

SCHEDULE AND FEEDS

Validates the exact 90-day window, phase boundaries, per-feed age, decimals and oracle risk settings.

ARTIFACTS

Rejects testnet inputs and strips signer and RPC secrets from public artifacts.

Wrong-Chain Protection

WRONG-CHAIN GUARD · PASS

CONSTRUCTOR GUARD

AagentStakingV1 refuses deployment outside BSC mainnet chain 56.

REGRESSION

Independent chain-97 configuration confirms the rejection path.

Operational Controls

KEEPER

Monitors only purchase-capable modes before sale end and fails closed on missing or malformed configuration.

PAUSE SEMANTICS

Purchase feed failures cannot disable redemption-only or post-launch redemption modes.

ROLE CONTROLS

Owner-only changes, pause requirements and reserved-asset withdrawal limits are explicitly tested.

REGRESSION MATRIX

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Regression Matrix

ACCOUNTING

Cost basis, partial refunds, aggregate liability, reserve withdrawals and inventory claims.

ORACLE

Stale, future, incomplete, out-of-bound and excessive-deviation rounds.

CUSTODY

Reentrancy, fee-on-transfer tokens, nonce replay and duplicate claims.

DEPLOYMENT CHECKLIST

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Deployment Checklist

BYTECODE

Deploy only byte-identical code bound to the printed scope root and verify it publicly.

CONFIGURATION

Bind registry endpoints, production feeds, beneficiary, owner roles and exact sale window.

FUNDING

Fund AGNT inventory and refund reserves before enabling purchase-capable modes.

MAINNET ACCEPTANCE

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Mainnet Acceptance

MINIMUM TRANSACTION

Execute a controlled funded wallet purchase, redemption and staking acceptance flow.

EVIDENCE

Retain transaction hash, receipt, input, value, events, confirmations and canonical block hash.

DECISION

Activation is an operational release decision after these gates, not an unresolved source-code finding.

CONCLUSION

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Conclusion

ASSESSMENT COMPLETE · ALL TRACKED CODE FINDINGS RESOLVED

CODE STATUS

All nine tracked findings are resolved in the current source snapshot.

VERIFICATION STATUS

Compilation, targeted tests, wrong-chain test, coverage, static analysis and production dependency audit completed.

RELEASE STATUS

Mainnet activation remains conditional on the printed deployment and funded-wallet acceptance gates.

DISCLAIMER

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Disclaimer

TECHNICAL SCOPE

This document records a point-in-time smart-contract security assessment of the identified source.

NO INVESTMENT ADVICE

Nothing in this document is financial, legal, tax or investment advice.

NO ABSOLUTE ASSURANCE

Testing and analysis cannot guarantee that all vulnerabilities are absent or that future changes remain secure.
